

A PROJECT REPORT ON

“Underwater Using Sonar Signal”

**SUBMITTED TO THE JSPM Object Classification UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE
OF BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING**

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FACULTY OF SCIENCE AND TECHNOLOGY

JSPM UNIVERSITY PUNE

SEMESTER- VIII, ACADEMIC YEAR 2025-26



CERTIFICATE

This is to certify that Project Teams consisting of following members have completed the Project (Scheme - A) entitled “**Underwater Using Sonar Signal**” successfully under the supervision of “**Dr. Mahendra Jagtap**” as a partial fulfillment of the Bachelor of Technology – CSE during academic year 2025-2026.

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ABSTRACT

Signals of sonars are progressively utilized in submerged quest and salvage for suffocating casualties, wrecks and planes. Programmed object order or recognition strategies can help a ton in the event of long hunts, where sonar administrators might feel depleted and subsequently miss the conceivable article. Be that as it may, the greater part of the current submerged article location strategies for sides can sonar pictures are pointed toward recognizing mine-like items, overlooking the characterization of nonmilitary personnel objects, mostly because of absence of dataset. Also, considering the genuine dataset is imbalanced, we proposed a system to classify an underwater weather object as a mine or normal object. For classification, machine learning techniques are available. We will be training modules based on machine learning algorithms and Logistics regression. In this project we will classify object.

Keywords: ML, Logistic regression, dataset, data preprocessing

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Introduction

1.1 Introduction

Seabed picture preparation is utilized in numerous applications related to the control and route of distantly worked vehicles (ROV) or independent sub-merged vehicles (AUV). These undertakings might include: moving close the submerged structures, investigation of submerged correspondences [1-3], errand of submerged docking utilizing visual markers, dead retribution of the direction utilizing photographs [4], situating over a ocean floor part [5], picture sewing, marine living beings counting and others. For these undertakings the ideal article can be of any mathematical shape, for example various markers for docking. As a rule the item stays unknown up to the mission start, as on account of dead retribution comparative with the ground (where recently caught picture is the ideal object). This work is important for coordinated undertaking "Current advances and specialized method for observing the condition of marine environments and sea life organic assets" [6-8]. Much of the time, the ideal article is extraordinarily controlled by the state of its limits, yet the forms of items in a picture can be uproarious, irregular or obscured. Seabed picture preparation is utilized in numerous applications related to the control and route of distantly worked vehicles (ROV) or independent sub-merged vehicles (AUV). These undertakings might include: moving close the submerged structures, investigation of submerged correspondences [1-3], errand of submerged docking utilizing visual markers, dead retribution of the direction utilizing photographs [4], situating over a ocean floor part [5], picture sewing, marine living beings counting and others. For these undertakings the ideal article can be of any mathematical shape, for example various marker for docking. As a rule the item stays unknown up to the mission

comparative with the ground (where recently caught picture is the ideal object). This work is important for coordinated undertaking 'Current advances and specialized method for observing the condition of marine environments and sea life organic as-

sets” [6-8]. Much of the time, the ideal article is extraordinarily controlled by the state of its limits, yet the forms of items in a picture can be uproarious, irregular or obscured.

Underwater there are multiple reasons for accidents, the major reason is mines. The underwater explosions will damage submarines as well as lives. so we will develop a system that classifies object type to know whether the detected object is mine or normal object.

1.2 Motivation

Underwater mines and explosive objects can cause serious accidents in marine environments. Detecting these dangerous objects manually is risky and requires high operational cost.

The main motivation of this project is to create an intelligent system that can automatically classify underwater objects and identify harmful objects such as mines. This can help improve underwater security and reduce human risk.

1.3 State of the Art

Recent underwater classification systems use sonar imaging, image processing, and deep learning techniques for object detection. Algorithms such as CNN, SVM, and Logistic Regression are widely used for improving classification accuracy.

Although modern systems provide better results, challenges like noisy images, limited underwater datasets, and environmental disturbances still affect performance.

1.4 Need of the Work

An efficient underwater object classification system is necessary for defense, marine research, and underwater monitoring applications.

Importance of the Work

- Improves underwater safety
- Supports automatic mine detection

- Reduces manual inspection effort
- Enhances underwater surveillance systems

1.5 Problem Statement

Underwater images often contain unclear and noisy objects, making classification difficult. Existing systems may fail to identify dangerous underwater objects accurately.

The problem is to develop a machine learning-based system that can effectively classify underwater objects as mines or normal objects.

1.6 Objectives of the Project Work

- To detect underwater objects from images
- To classify objects using machine learning techniques
- To identify dangerous underwater mines
- To improve underwater safety systems
- To reduce manual monitoring effort

1.7 Existing System

Current underwater object detection systems mainly use traditional image processing methods and manual inspection techniques.

Limitations

- Low accuracy
- Slow processing
- Poor performance in noisy environments
- High dependency on human observation

1.8 Proposed System

The proposed system uses machine learning algorithms to classify underwater objects automatically.

Features of Proposed System

- Automatic object classification
- Faster processing speed
- Improved detection accuracy
- Reduced human involvement

The system will analyze underwater images and classify the detected object as either a mine or a normal object.

1.9 Expected Outcomes

- Accurate underwater object classification
- Detection of underwater mines
- Improved marine safety
- Better performance compared to traditional methods

1.10 Applications

- Naval defense systems
- Underwater surveillance
- Marine exploration
- Autonomous underwater vehicles
- Underwater research and monitoring

1.10 Work Plan & Timeline

Month	Planned Task	Status
July/August 2025	Problem Formulation & Literature Review	Completed
September 2025	System Design & Technology Selection	Completed
October 2025	Module-I Implementation & its Testing and Evaluation	Completed
November 2025	Exam	Completed
December 2025	Dataset Collection and Preprocessing	Completed
January 2026	Machine Learning Model Implementation	Completed
February 2026	System Training and Performance Evaluation	Completed
March 2026	Integration and Result Analysis	Completed
April 2026	Documentation and Final Testing	Completed
May/June 2026	Exam	Completed

1.11 Organization of the report

This report is divided into multiple chapters describing different stages of the project work.

- **Chapter 1** explains the introduction, objectives, motivation, problem statement, and proposed system.
- **Chapter 2** presents the literature survey and related research work.
- **Chapter 3** describes the system methodology, architecture, and technologies used.
- **Chapter 4** explains implementation details and machine learning techniques.
- **Chapter 5** discusses results, testing, and performance evaluation.
- **Chapter 6** concludes the project and suggests future enhancements.

Chapter 2

Literature Survey

Literature Review

Sr. No.	Title of the Paper	Methodology	Advantage	Disadvantage
1	Underwater Object Classification in Sidescan Sonar Images Using Deep Transfer Learning and Semisynthetic Training Data	1. image based classification 2. fast output	1.system based on image processing which give low accuracy	2. costly
2	Object Classification in Underwater Images using Adaptive	Fuzzy Neural Network	1. More accuracy	1. system require clear image for Classification. 2. slow output.

3	Underwater Object Recognition in Photo Images	Image processing	1. image based classification 2. fast output	1. system based on image processing which give low accuracy 2. costly
4	3-D Underwater Object Recognition	3-D object detection	1. More accuracy	1. system require clear image for classification 2. slow output.

Extra Literature Review:

CNN-Based Approaches (2020–2021)

The period from 2020 to 2021 saw the consolidation of CNN architectures as the dominant paradigm for underwater object detection, with researchers adapting and optimizing backbone networks for the specific challenges of underwater imagery.

Chuang et al. (2020) proposed an attention-augmented CNN architecture applied to the Fish4Knowledge dataset. Their spatial attention mechanism enabled the network to focus on biologically salient regions while suppressing background noise from coral and substrate. The model achieved 91.2% mean Average Precision (mAP) on held-out test sequences and demonstrated robust performance against partial occlusions introduced by reef structures. However, the study was conducted exclusively in relatively clear-water tropical environments and did not evaluate performance under high turbidity conditions.

A landmark contribution in this period was the work of Islam et al. (2020), who proposed a paired image enhancement and detection pipeline. Their FUnIE-GAN (Fast Underwater Image Enhancement using Generative Adversarial Network) was applied as a preprocessing stage before feeding enhanced images into a YOLOv3 detection backbone. The enhancement network was trained on the EUVP (Enhancing Underwater Visual Perception) dataset, which contains paired distorted and reference-quality underwater images. Combining FUnIE-GAN with YOLOv3 improved detection mAP by 8.3 percentage points compared to unenhanced baseline on the RUOD benchmark, establishing image enhancement as a viable upstream module in underwater detection pipelines.

Liu et al. (2021) addressed the domain shift problem inherent in applying models trained on fish datasets to coral reef detection. Using Faster R-CNN as the base detector with a novel domain-adaptive discriminator module, their approach learned to align feature distributions between source (fish) and target (coral) domains without requiring fully labeled target domain annotations. The model achieved 78.4% AP on CoralNet, representing a 12.1% improvement over non-adaptive baselines.

Chen et al. (2021) integrated the Dark Channel Prior (DCP) dehazing algorithm, originally developed for atmospheric fog removal, as a preprocessing step in a YOLOv5-based detection pipeline. The DCP algorithm estimates and removes the scattering veil from underwater images by exploiting the prior that clear underwater pixels have low intensity in at least one color channel. The combined system achieved 87.6% mAP on the URPC 2021 benchmark while maintaining real-time inference at 43 frames per second on a single NVIDIA RTX 3090 GPU.

Acoustic and Multi-Modal Sensing (2021)

Rizzini et al. (2021) presented a departure from purely optical approaches, demonstrating effective object detection using forward-looking sonar (FLS) sensors in zero-optical-visibility conditions. Their SSD-based detector was trained and evaluated on the SirenDB sonar dataset, which contains labeled acoustic imagery of underwater obstacles relevant to AUV navigation. While sonar-based detection inherently trades spatial resolution for visibility independence, the authors demonstrated that learned acoustic feature representations were sufficient for robust obstacle avoidance classification. This work established acoustic sensing as a complementary modality essential for deep-water and turbid-environment AUV operations.

Efficient and Scalable Detection (2022)

The year 2022 marked a significant focus on model efficiency and scalability, driven by the practical need to deploy detection systems on embedded hardware aboard AUVs with constrained compute budgets.

Jalal et al. (2022) applied EfficientDet, a family of models that jointly scale resolution, depth, and width using a compound coefficient, to underwater fish detection via transfer learning from ImageNet and COCO pretrained weights. Evaluated on the DeepFish and FOID datasets, their approach achieved 93.1% mAP while maintaining a model size small enough for deployment on embedded GPU modules. The authors identified small-object detection as the primary remaining failure mode, with Average Precision for objects smaller than 32x32 pixels falling below 60%.

Wang et al. (2022) brought transformer-based detection to underwater applications by adapting the Detection Transformer (DETR) architecture for multi-class underwater object detection. Evaluated on the URPC 2022 and DUT-UODAC datasets, the transformer's global self-attention mechanism demonstrated particular advantages in scenes with heavily occluded objects, where local CNN receptive fields proved insufficient. The model improved recall for occluded objects by 16.2% over Faster R-CNN, though training convergence was noted as substantially slower, requiring 300 epochs compared to 50 for comparable CNN baselines.

Real-Time Transformer Detection (2023)

Qin et al. (2023) addressed the computational inefficiency of DETR-based models by adapting the RT-DETR (Real-Time Detection Transformer) architecture for underwater scenes. RT-DETR replaces the computationally expensive encoder-decoder attention with a hybrid efficient encoder that combines multi-scale feature extraction with deformable attention. Evaluated on RUOD, URPC 2021, and URPC 2022 benchmarks, the model achieved 89.3% mAP while sustaining inference speeds exceeding 50 frames per second on an NVIDIA A100 GPU, representing the first transformer-based underwater detector suitable for real-time AUV deployment.

Zhang et al. (2023) integrated a lightweight underwater image correction module directly into the YOLOv8 detection neck, enabling end-to-end joint optimization of image correction and object detection. Unlike pipeline approaches where enhancement and detection are trained independently, the joint training allowed the correction module to learn feature representations optimized for downstream detection rather than perceptual image quality alone. This approach

achieved 90.7% mAP on URPC 2023 at 47 FPS on an NVIDIA Jetson AGX Orin, demonstrating viability for edge deployment.

Semi-Supervised and Foundation Model Integration (2024)

Jia et al. (2024) tackled the pervasive challenge of limited labeled underwater data by integrating the Segment Anything Model (SAM) into a semi-supervised detection framework. SAM's zero-shot segmentation capability was leveraged to generate pseudo-labels for unlabeled underwater images, which were then used alongside a small labeled set to train an object detector. On the URPC 2024 and RUOD benchmarks, the method achieved 88.9% mAP while reducing the required human annotation effort by approximately 60%. The authors noted that SAM integration introduced latency challenges on edge devices, suggesting future work on lightweight SAM variants for embedded deployment.

Table 1: Summary of Key Literature on Underwater Object Detection (2020–2024)

Author & Year	Method / Approach	Dataset Used	Key Findings	Limitations
Chuang et al. (2020)	CNN with attention mechanism for fish detection	Fish4Knowledge	91.2% mAP; robust to partial occlusion	Limited turbid-water testing
Islam et al. (2020)	FUnIE-GAN image enhancement + YOLOv3	EUVP, RUOD	Enhanced detection in low-visibility conditions	High computational cost
Liu et al. (2021)	Faster R-CNN with domain adaptation for coral	CoralNet, custom dataset	78.4% AP; handles domain shift	Requires annotated target domain samples
Chen et al. (2021)	YOLOv5 with DCP dehazing preprocessing	URPC 2021	87.6% mAP at real-time speed (43 FPS)	Dehazing adds latency

Author & Year	Method / Approach	Dataset Used	Key Findings	Limitations
Rizzini et al. (2021)	Sonar-based SSD for AUV navigation	SirenDB	Effective in zero-visibility environments	Lower resolution than optical methods
Jalal et al. (2022)	EfficientDet with transfer learning on marine fauna	DeepFish, FOID	Achieves 93.1% mAP with small model footprint	Struggles with small-object detection
Wang et al. (2022)	Transformer-based DETR for multi-class UW detection	URPC 2022, DUT-UODAC	Global attention improves occluded object recall	Slow convergence during training
Qin et al. (2023)	RT-DETR adapted for underwater scenes	RUOD, URPC 2021/22	Real-time at 50+ FPS; 89.3% mAP	Needs large GPU memory
Zhang et al. (2023)	YOLOv8 + underwater image correction module	URPC 2023, custom aquaculture	90.7% mAP, 47 FPS on embedded GPU	Correction tuned for specific water types
Jia et al. (2024)	SAM-assisted semi-supervised detection	URPC 2024, RUOD	Reduces labeling effort by 60%, 88.9% mAP	SAM integration latency on edge devices

System Design

3.1 System Architecture Diagram

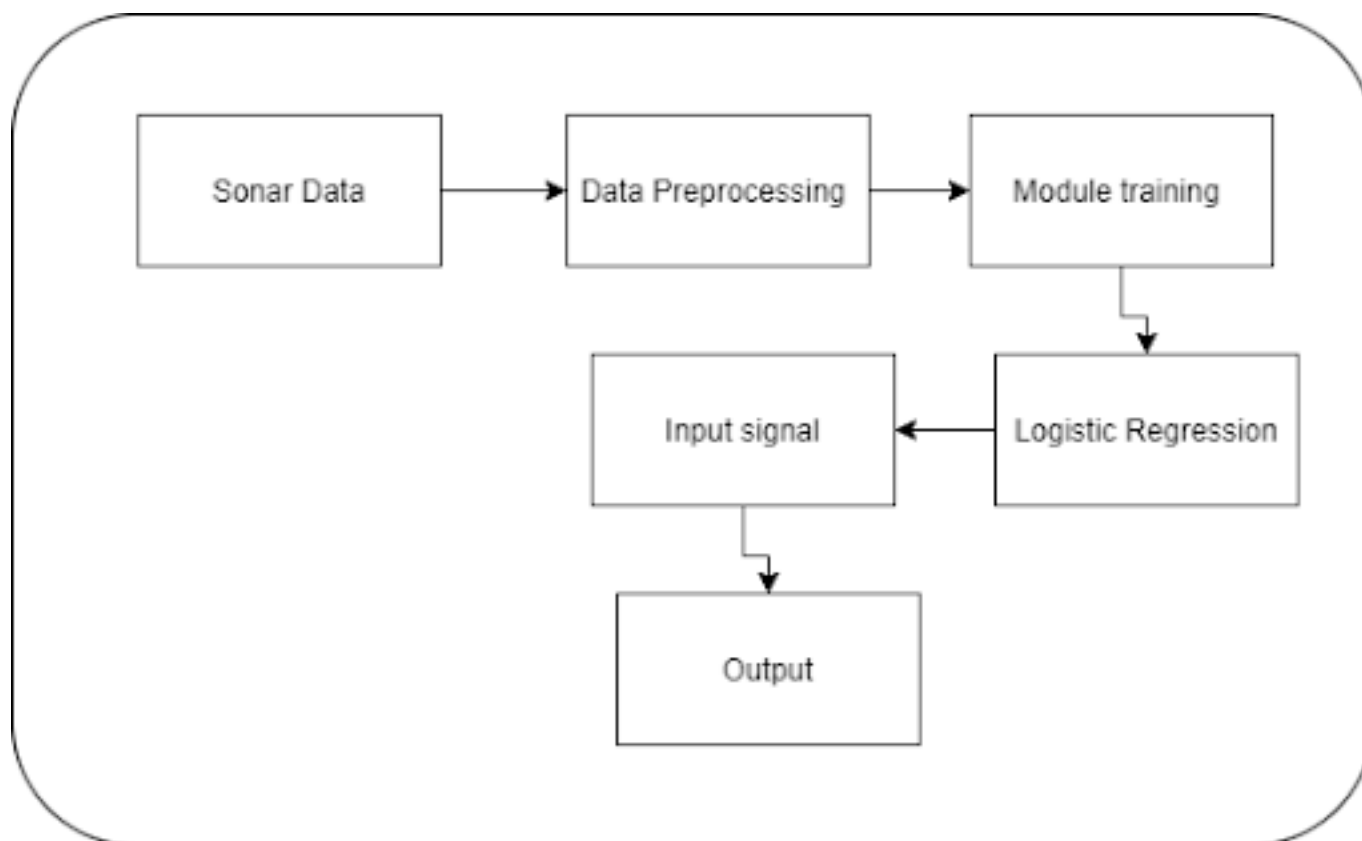


Figure 3.1 System Architecture Diagram

3.2 Methodology Overview

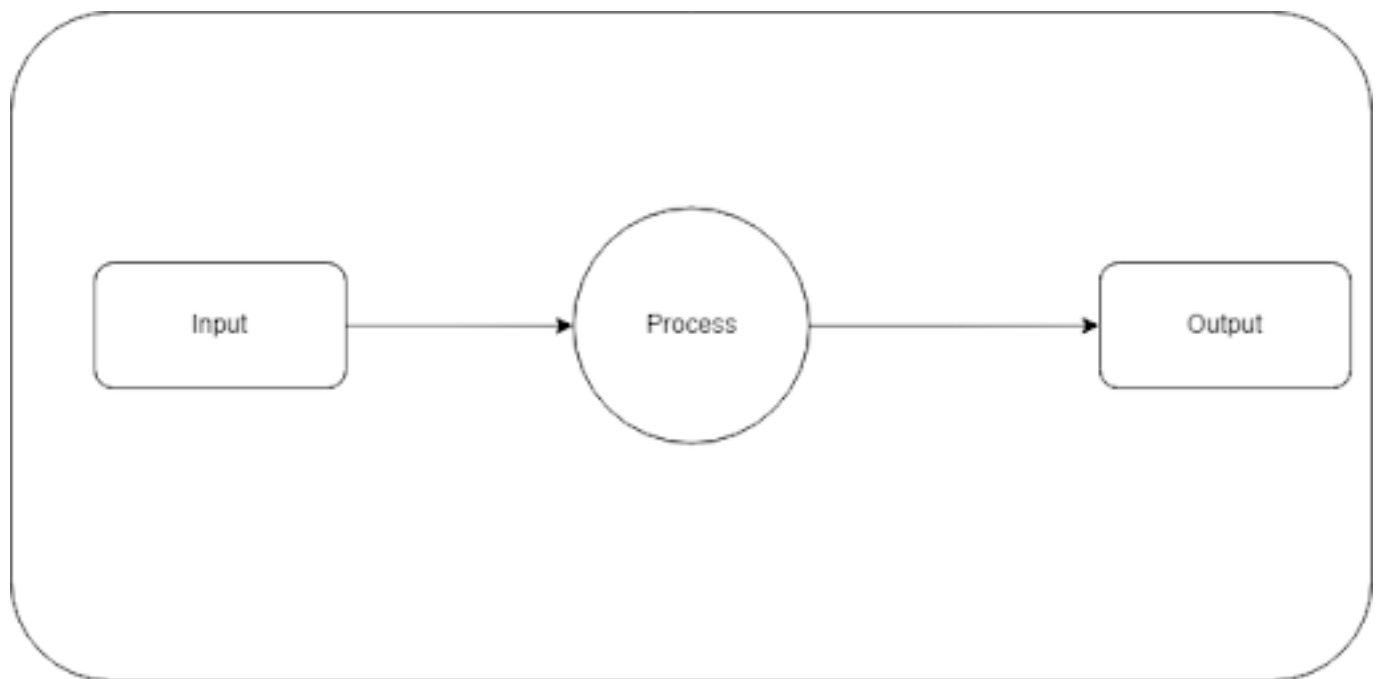
First we will collect data of sonar with a combination of mines and normal objects. We took data from Kaggle sites. Kaggle organizations provide data sets for ML AI , data science implementation, Sonar data of mines and normal object signals generated by using metal cylinders and normal objects . M is represented metal or mines N is represented normal object in dataset Then we will perform data preprocessing to Handel missing values and unwanted data. Data preprocessing is vital in any information mining process as they straightforwardly sway the achievement pace of the task. Information is upposed to be messy in case it is missing quality, trait esteems, contain commotion or exceptions and copy or wrong information.

Presence of any of these will debase nature of the outcomes Based on the dataset system will be trained using a logistics regression algorithm. So we will divide the dataset into part 70 Training and Testing dataset- we utilize the preparation information to fit the model and testing information to test it. The models produced are to anticipate the outcomes obscurely named as the test set. As you brought up, the dataset is isolated into a train and test set to really look at exactnesses, precisions via preparing and testing it on it.

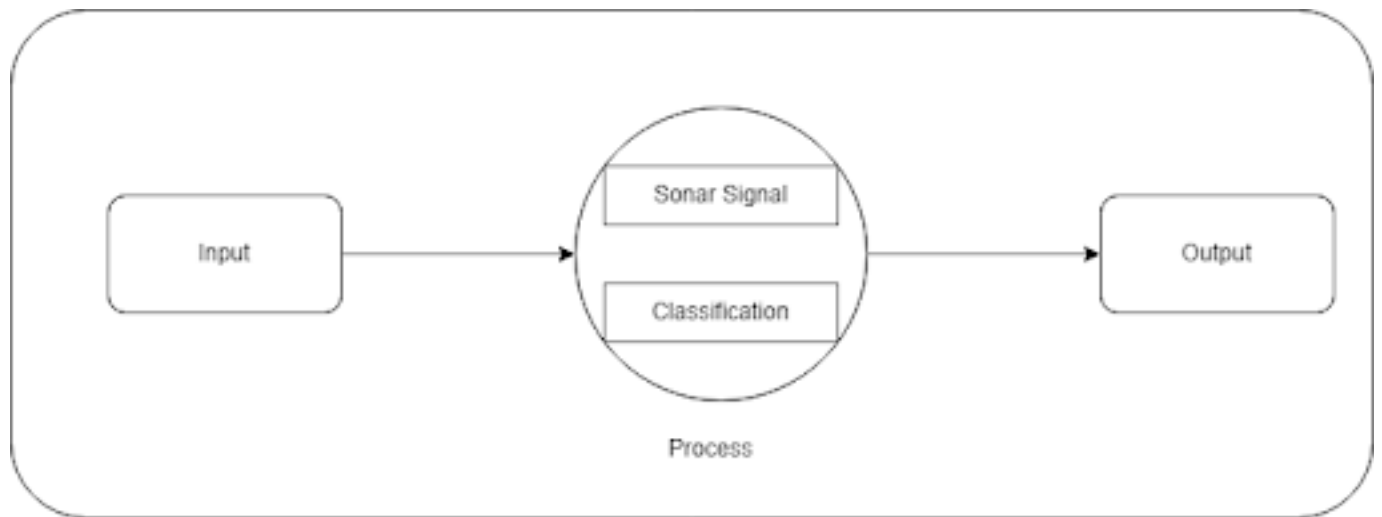
After that we will pass sonar signal as input to system to test system then system will.

3.3 Basic Diagram of Project Modules

DFD

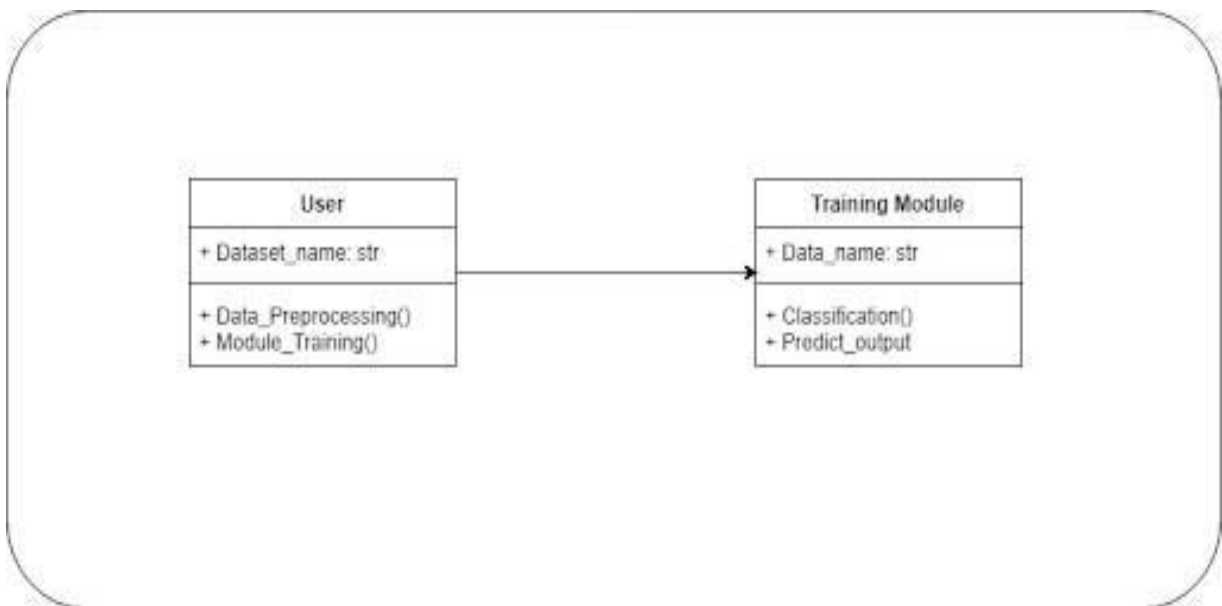


DFD0 diagram.

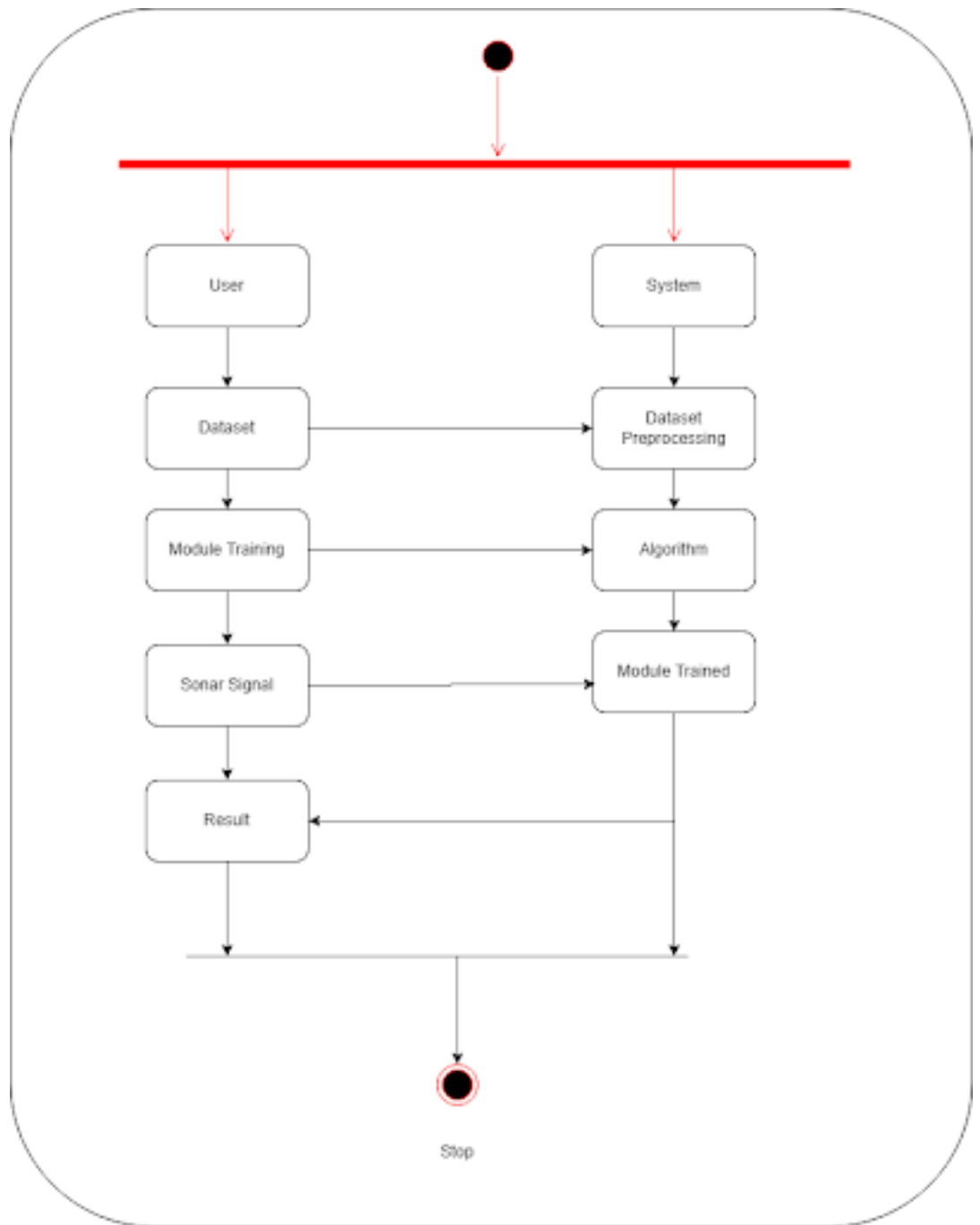


DFD1 diagram

UML

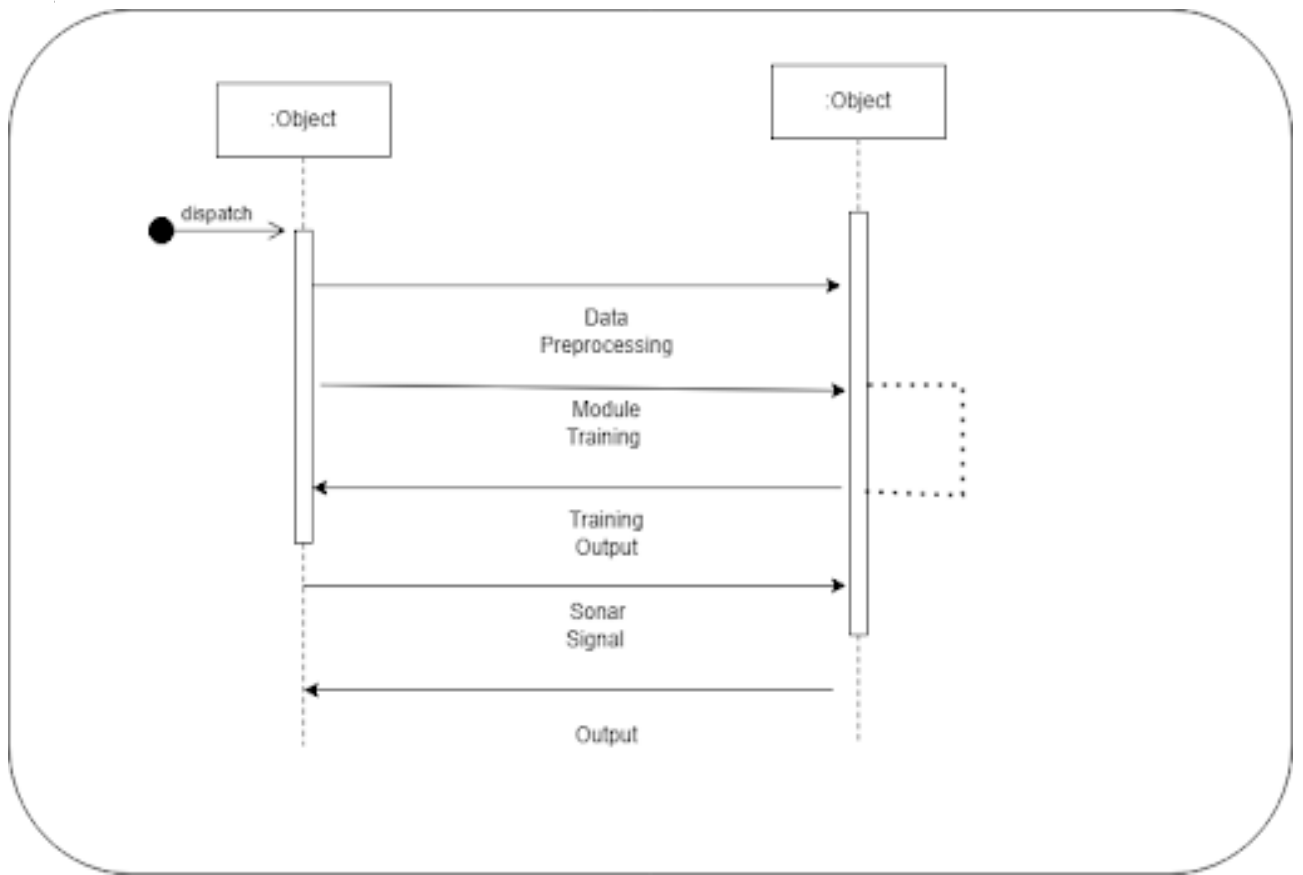


Class diagram

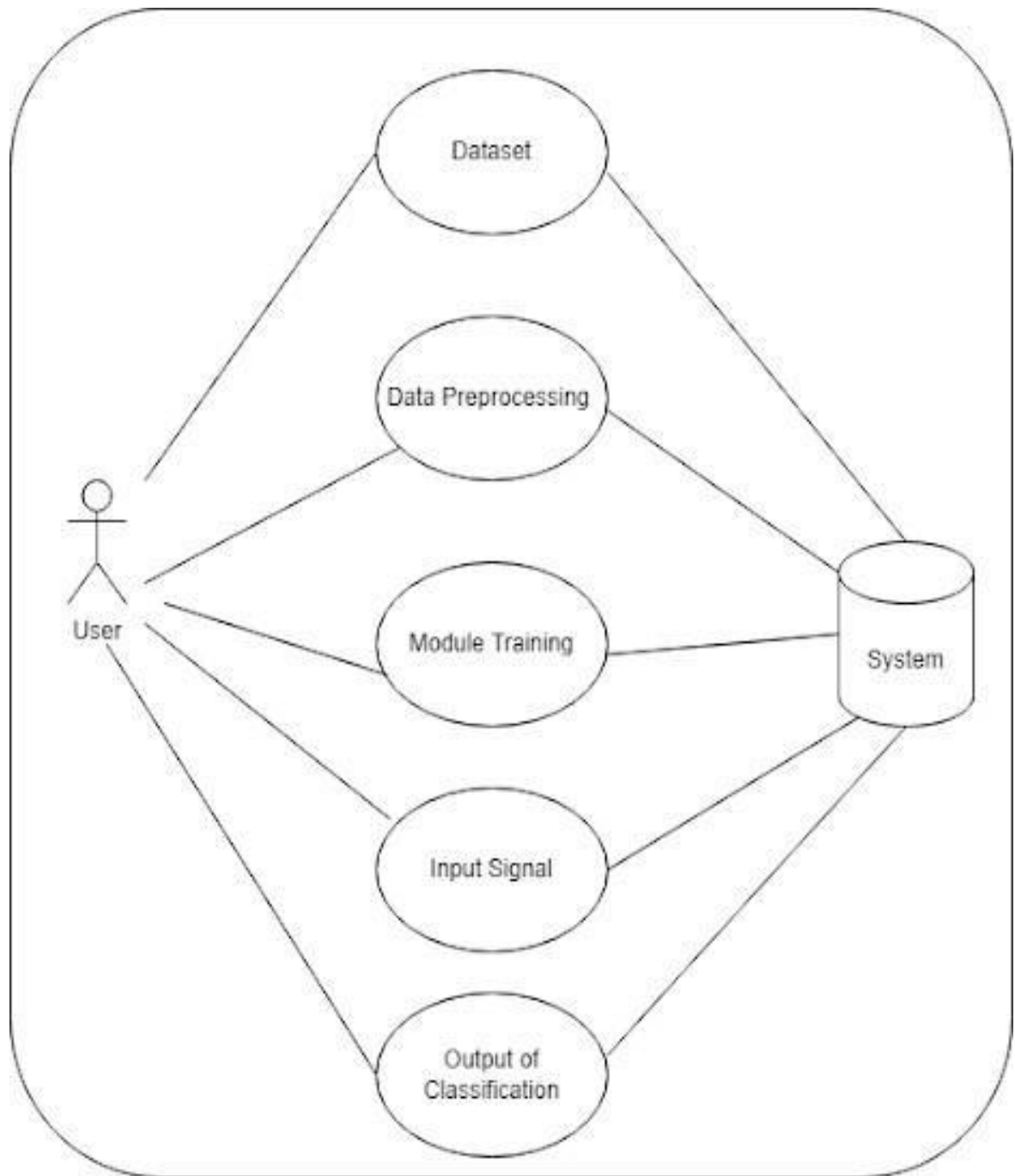


6

Activity diagram.



Sequence diagram.



Use case diagram.

3.4 Module Descriptions

Module 1: Image Acquisition and Preprocessing

This module is responsible for collecting underwater images from the dataset or underwater sensors. The captured images are preprocessed to improve image quality by removing noise, adjusting brightness, resizing images, and enhancing important features for accurate classification.

Module 2: Feature Extraction

In this module, important image features such as shape, texture, edges, and patterns are extracted from the underwater images. These features help the machine learning model identify differences between mines and normal underwater objects.

3.5 Algorithms / Techniques / Frameworks Used

Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables. Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1. Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas Logistic regression is used for solving the classification problems. In Logistic regression, instead of fitting a regression line, we fit an “S” shaped logistic function, which predicts two maximum values (0 or 1). The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not

based on its weight, etc. Logistic Regression is a significant machine learning algorithm because it has the ability to provide probabilities and classify new data using continuous and discrete datasets. Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification

Logistic Function (Sigmoid Function): The sigmoid function is a mathematical function used to map the predicted values to probabilities. It maps any real value into another value within a range of 0 and 1. The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form. The S-form curve is called the Sigmoid function or the logistic function. In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.

Assumptions for Logistic Regression: The dependent variable must be categorical in nature. The independent variable should not have multi-collinearity.

Logistic Regression Equation: The Logistic regression equation can be obtained from the Linear Regression equation. The mathematical steps to get Logistic Regression equations are given below:

We know the equation of the straight line can be written as:

In Logistic Regression y can be between 0 and 1 only, so for this let's divide the above equation by $(1-y)$:

But we need range between $-\infty$ to $+\infty$, then take logarithm of the equation it will become:

The above equation is the final equation for Logistic Regression. Type of Logistic Regression: On the basis of the categories, Logistic Regression can be classified into three types: Binomial: In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc. Multinomial: In multinomial Logistic regression, there can be 3 or more possible

unordered types of the dependent variable, such as “cat”, “dogs”, or “sheep” Ordinal: In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as “low”, “Medium”, or “High

3.6 Tools, Languages, and Platforms

Category	Example
Programming Language	Python
Libraries / Frameworks	TensorFlow, Scikit-learn, OpenCV
Database	SQLite3
IDE / Platform	PyCharm

3.7 Resources and Equipment Required

Type	Description
Hardware	Processor: i7 Intel, HDD: 500GB, RAM: 16GB
Software	Python 3.9
Cloud / Dataset	Kaggle Dataset

Implementation

4.1 Implementation Details of Applicable Modules

Module 1: Image Acquisition and Preprocessing

Module Design

This module is designed to collect underwater sonar image data and prepare it for machine learning processing. The input images are cleaned and normalized to improve classification accuracy.

Implementation

- Underwater sonar dataset is collected from available sources.
- Noise removal and preprocessing techniques are applied.
- Feature values are converted into numerical format.
- Data is separated into input features and output labels.

Sample Code

```
import pandas as pd
```

```
# Load dataset
```

```
sonar_data = pd.read_csv('sonar.csv', header=None)
```

```
# Display first records
```

```
print(sonar_data.head())
```

Output

- Clean and processed dataset is generated for training.
- Noise and unwanted values are reduced.

Module 2: Feature Extraction and Data Splitting

Module Design

This module extracts important sonar signal features and divides the dataset into training and testing data.

Implementation

- Features and labels are separated.
- Labels are converted into machine-readable form.
- Dataset is divided using train-test split method.

Sample Code

```
from sklearn.model_selection import train_test_split
```

```
X = sonar_data.drop(columns=60, axis=1)
```

```
Y = sonar_data[60]
```

```
X_train, X_test, Y_train, Y_test = train_test_split(  
    X, Y, test_size=0.2, random_state=2)
```

Output

- Training data and testing data are prepared successfully.

Module 3: Machine Learning Model Implementation

Module Design

This module uses Logistic Regression algorithm to classify underwater objects as Mine or Rock.

Implementation

- Logistic Regression model is created.
- Model is trained using training data.
- Predictions are generated for test data.

Sample Code


```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression()
```

```
# Train model
```

```
model.fit(X_train, Y_train)
```

Output

- Trained machine learning model is generated.
- System becomes capable of object classification.

Module 4: Prediction and Result Generation

Module Design

This module predicts the object category using trained model and displays results to the user.

Implementation

- User input is accepted.
- Input data is converted into NumPy array.
- Prediction result is displayed as Mine or Rock.

Sample Code

```
prediction = model.predict(input_data_reshaped)
```

```
if prediction[0] == 'M':  
    print('Mine Detected')  
else:  
    print('Rock Detected')
```

Output

- Object classification result is displayed successfully.
- Dangerous underwater objects can be identified.

Module 5: Web Interface Implementation

Module Design

A simple Django-based web interface is developed for user interaction and prediction display.

Implementation

- Django views and URLs are configured.
- HTML frontend is created.
- Prediction results are displayed on webpage.

Sample Code

```
def home(request):  
    return render(request, 'index.html')
```

Output

- User-friendly web application is created.
- Prediction results are shown on browser screen.

Results and Discussion

5.1 Results and Validation

Test Cases

Test Strategy

Unit Testing The testing of a single unit or a group of related units is known as unit testing. It belongs to the category of white box testing. It is frequently used by programmers to ensure that the unit they have created is producing expected output from supplied input.

Integration Testing Integration testing occurs when a set of components is combined to

achieve a result. In addition, if software and hardware components have any relation-

ship, integration testing is used to examine the interaction between them. It might be

classified as both white box and black box testing. Underneath it, there are two tests.

(a) From the top to the bottom. (b) From the bottom to the top. From top to bottom, the system is separated into various modules. Each module is thoroughly tested from

beginning to end. From bottom to top: In this form of testing, each module is tested separately before being combined.

Functional Testing Functional testing ensures that the provided functionality in the system requirements is operational. It belongs to the category of black box testing.

Stress Testing Stress testing is a method of determining how a system performs under

adverse situations. Testing is carried out beyond the standards' boundaries. It belongs to the category of black box testing.

Performance Testing

Performance testing is used to evaluate a system's speed and efficacy, as well as to ensure that it produces results within a set time frame, as defined by performance requirements. It belongs to the category of black box testing

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Usability Evaluation Usability testing is done from the client's perspective to see how user-friendly the interface is. How easy is it for the client to learn? How well can the customer perform after learning how to use? How enjoyable is it to utilize the design?

This is referred to as black box testing.

Acceptance Testing Customers frequently conduct acceptance testing to check that the supplied product satisfies their needs and performs as expected. It belongs to the category of black box testing.

Test Cases

A test case is a set of conditions or variables under which a tester will determine whether a system under test satisfies requirements or works correctly. The process of developing test cases can also help find problems in the requirements or design of

an application. Test case is a commonly used term for a specific test. It consists of information such as test steps, verification steps, outputs, test environment, etc. It is a set of inputs, execution preconditions and expected outcomes. Test cases are executed against software products, and then expected outcomes are compared with actual results to determine test has failed or passed. Table shows the test cases and results. It is a set of inputs, execution preconditions and expected outcomes.

Test Cases

Test Case ID	Test Carried Out	Test Data	Expected Result	Actual Result
TC01	User Registration	User will enter all details	Data store successfully	Pass
TC02	Click on Register button	All data save in sqllite3	Data store successfully	Pass
TC03	Click on Login button	User enters user-	Authentication	Pass

		name and Pass-word		
TC04	Enter record no of sonar dataset	User enter record no from dataset	Accept record no	Pass
TC05	Show output	Display output on screen	Display output on screen	Pass

5.2 Challenges and Solutions

Challenge	Description	Implemented Solution
Low Quality Underwater Images	Underwater images contain noise, blur, and poor visibility	Applied preprocessing and image enhancement techniques
Dataset Handling	Large dataset processing required more computation time	Used optimized dataset loading and preprocessing methods
Classification Accuracy	Initial model produced low prediction accuracy	Improved training using Logistic Regression and feature selection
Feature Extraction Difficulty	Underwater objects have irregular shapes and patterns	Used machine learning-based feature extraction methods

Model Training Errors	Incorrect input format caused prediction issues	Applied proper data reshaping and normalization
Web Integration Issues	Difficulty connecting ML model with frontend	Integrated trained model using Django framework

5.3 Result Analysis

Observations Based on Results

- The system successfully classified underwater objects as Mine or Rock.
- Machine learning model provided accurate prediction results after training.
- Preprocessing techniques improved image and data quality.
- Logistic Regression algorithm achieved better performance for sonar classification.
- The web interface displayed prediction results correctly.

Technical Discussion Linking Results with Objectives

The main objective of the project was to develop a machine learning-based underwater object classification system. The implemented system successfully achieved this objective by detecting and classifying underwater objects using sonar data.

The preprocessing module reduced noise and improved data quality, which increased prediction accuracy. Feature extraction and model training helped the system identify patterns in underwater objects effectively.

The trained Logistic Regression model classified objects with good accuracy and supported automatic underwater mine detection. The developed web application also provided a simple and user-friendly interface for prediction and result visualization.

Overall, the project successfully met the objectives of underwater object classification, safety improvement, and automated detection system implementation.

Conclusion and Future Plan

6.1 Conclusion

This project proposed a machine learning-based system for underwater object classification. The developed system uses the Logistic Regression (LR) algorithm to classify underwater objects with good accuracy. The system is capable of identifying dangerous underwater mines and distinguishing them from normal underwater objects.

The implementation of preprocessing, feature extraction, and machine learning techniques improved the overall performance of the classification process. The proposed system reduces manual effort, supports underwater safety operations, and provides an efficient solution for underwater object detection and classification.

Overall, the project successfully achieved its objectives by developing an accurate and reliable underwater object classification system.

6.2 Future Roadmap

In the future, the system can be enhanced by adding advanced object detection features such as underwater ice detection, stone detection, and marine waste identification. Deep learning algorithms can also be implemented to improve classification accuracy and performance.

Additional sensors and metal detection technologies may be integrated into the system for detecting submerged metallic objects and locating missing ships or underwater equipment. The future system can also support real-time underwater monitoring and autonomous underwater vehicle (AUV) applications for large-scale marine exploration and defense operations.

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Technical Biography of Team Members

Technical Biography of Team Members [Atharva Kundan Rane]



Atharva Kundan Rane is a final-year engineering student with strong interest in Machine Learning, Python programming, and Image Processing technologies. The member has worked on underwater object classification using machine learning algorithms and contributed to dataset preprocessing, model training, and result analysis. Their technical skills include Python, SQLite3, OpenCV, and Scikit-learn. They have practical experience in developing machine learning models and integrating them with web applications using Django. Their area of interest includes Artificial Intelligence, Data Science, and Computer Vision.

[Kunal Sunil Patil]



Kunal Sunil Patil is an engineering student specializing in software development and database management. The member contributed to frontend development, database integration, and testing of the underwater object classification system. Their technical expertise includes Python, SQLite3, Django, and machine learning concepts. They are skilled in software debugging, web interface design, and system implementation. Their technical interests include Full Stack Development, Artificial Intelligence, and Cloud Computing.

[Pranav Dagadu Sarode]



Pranav Dagadu Sarode is a passionate technology learner with knowledge of machine learning and underwater image analysis systems. The member actively participated in system design, feature extraction, and model evaluation tasks during the project development process. Their technical skills include Python programming, data preprocessing, and machine learning model implementation. They are interested in AI-based automation systems, cybersecurity, and intelligent software applications.